INCH-POUND
MIL-PRF-3098/120D
27 July 2005
SUPERSEDING
MIL-PRF-3098/120C
27 August 1997

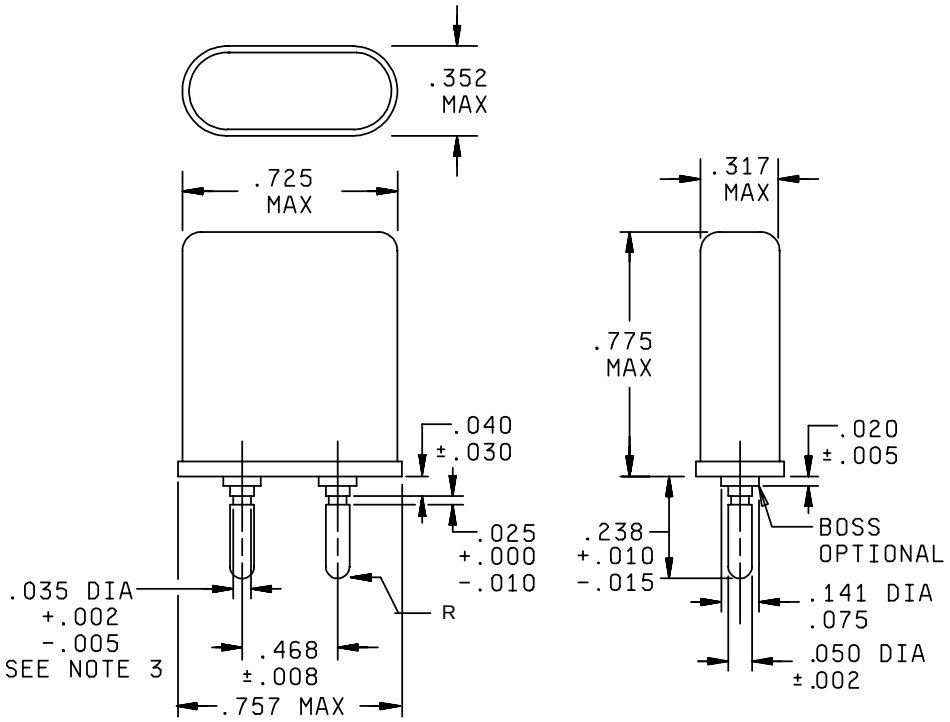
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR131/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; controlled; antiresonance.



INCHES	MM
.002	.05
.005	.13
.008	.20
.010	.25
.015	.38
.020	.51
.025	.64
.030	.76
.035	.89
.040	1.02
.050	1.27
.075	1.91
.141	3.58
.238	6.05
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR131/U.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Frequency stability: ±5 ppm.

Antiresonance, load capacitance: 30.0 pF ±0.5 pF.

Frequency and equivalent resistance: See table I.

Reference temperature: 75°C ±1°C.

Frequency tolerance: ±20 ppm.

TABLE I. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
0.80 to 0.85	630
0.85+ to 0.90	600
0.90+ to 1.00	580
1.00+ to 1.12	540
1.12+ to 1.25	490
1.25+ to 1.37	450
1.37+ to 1.50	410
1.50+ to 1.62	380
1.62+ to 1.75	330
1.75+ to 1.87	300
1.87+ to 2.00	290
2.00+ to 2.12	270
2.12+ to 2.25	250
2.25+ to 2.60	200
2.60+ to 3.00	150
3.00+ to 3.40	110
3.40+ to 3.75	90
3.75+ to 4.00	75
4.00+ to 5.00	60
5.00+ to 7.00	35
7.00+ to 10.00	24
10.00+ to 15.00	22
15.00+ to 20.00	20

Mode of oscillation: Fundamental.

Operable temperature range: -55°C to +70°C and +80°C to +90°C, inclusive.

Operating temperature range (controlled): +70°C to +80°C, inclusive.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.

Rated drive level: 1.0 mW, maximum.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse): 1/

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz ±15 percent.
2.0 MHz + to 20.0 MHz ±10 percent.

Vibration: Method 201, MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: 0.8 MHz to 2.0 MHz ±15 percent.
2.0+ MHz to 20.0 MHz ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: This crystal unit is identical to the CR27/U, except for a load capacitance rating of 30 pF ±0.5 pF. This unit is not interchangeable with the CR27/U.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

1/ The violation of $P = I^2R$ is intentional at the higher resistance of the crystal unit.

Custodians:
Army - CR
DLA - CC

Review activities:
Army – AR, MI

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0811-15)

* NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:56Z
Check the source to verify that this is the current version before use.